Experiment on Large Language Model (LLM) System Prompting Bias

This document serves as evidence of an experiment demonstrating how an LLM's system prompt influences the factual information presented, even when querying about the same individual.

Purpose of the Experiment

To investigate the effect of specialized system instructions (system prompts) on the output of a Large Language Model (LLM) when asked to provide information about a specific person, **Professor Mohammad Gulrez**.

Experimental Data: Prompting and Responses

The LLM was given a different specialized system prompt for each test, followed by the same query: "Who is Professor Mohammad Gulrez?"

Test 1: System Prompt Focused on AI

System Prompt (Simulated): "You are an AI assistant specialized in Artificial Intelligence."

```
prompt = ChatPromptTemplate.from_messages([
    ("system", "You are an AI assistant specialized in Artificial Intelligence."),
    ("human", "{question}")
])

chain = prompt | llm
```

Query: "Who is Professor Mohammad Gulrez?"

```
response = chain.invoke({"question": "who is Professor Mohammad Gulrez in amu"})

print(response)
print(response.content)
```

(variable) response: AIMessage

LLM Response:

"As of my last knowledge update in October 2023, Professor Mohammad Gulrez is a faculty member at Aligarh Muslim University (AMU) in India. **He is known for his work in the field of artificial intelligence and computer science.** However, specific details about his research, publications, or current role may not be fully up-to-date. For the most accurate and current information, I recommend checking the official AMU website or relevant academic publications."

Test 2: System Prompt Focused on Ocean Studies

System Prompt (Simulated): "You are an AI assistant specialized in ocean studies"

```
prompt = ChatPromptTemplate.from_messages([
    ("system", "You are an AI assistant specialized in ocean studies."),
    ("human", "{question}")
])
```

Query: "Who is Professor Mohammad Gulrez?"

```
response = chain.invoke({"question": "who is Professor Mohammad Gulrez in amu"})

print("\nAnswer:\n")
print(response.content)
```

LLM Response:

"As of my last knowledge update in October 2023, Professor Mohammad Gulrez is a faculty member at Aligarh Muslim University (AMU) in India. **He is known for his contributions to research and education, particularly in the fields of ocean studies, marine biology, or related disciplines.** However, specific details about his research interests, publications, or current projects may not be readily available in my training data. For the most accurate and up-to-date information, I recommend checking AMU's official website or academic publications related to his work."

Observation and Conclusion

The experiment clearly demonstrates the significant biasing effect of the system prompt on the LLM's output. When prompted with a specialization, the model actively selected and

emphasized information about Professor Gulrez that aligned with that specialization (AI vs. Ocean Studies), despite being asked the identical neutral query.

This highlights the critical role of pre-contextualization (system prompting) in shaping the factual content and focus of an LLM's response, even potentially overriding or selectively presenting information from its knowledge base.